

Improved Probabilistic Star-Galaxy Separation Method for Rubin LSST Catalogs

ŽELJKO IVEZIĆ ¹, JINSHUO ZHANG ¹, NIRAV NEMALI,¹ DAN TARANU,² COLIN T. SLATER,¹
 ROBERT H. LUPTON,² AND OTHER RUBIN COAUTHORS³

¹*Department of Astronomy, University of Washington, Box 351580, Seattle, WA 98195, USA*

²*Department of Astrophysical Sciences, Princeton University, Princeton, NJ 08544, USA*

³*Rubin Observatory*

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ABSTRACT

The morphological classification of resolved and unresolved sources, commonly referred to as “star-galaxy separation”, is a fundamental step in the analysis of wide-field imaging surveys such as Rubin Observatory’s Legacy Survey of Space and Time (LSST). Star-galaxy separation becomes increasingly challenging at the faint magnitudes probed by LSST because galaxy counts rise rapidly with magnitude while their angular sizes become progressively smaller. LSST photometric catalogs provide a binary classifier **extendedness** in each band, based on the threshold difference between PSF, m_{PSF} , and CModel, m_{CModel} , magnitudes. While this parameter provides a useful baseline for morphological classification, its binary nature prevents the optimal combination of information from multiple bands, as well as from other classification methods (e.g., those based on colors or external datasets). In this work, we implement the Bayesian star-galaxy classification method of Slater, Ivezić & Lupton (2020), which overcomes these shortcomings. In each band, we model the distribution of sources in the two-dimensional space defined by $\Delta m = m_{\text{PSF}} - m_{\text{CModel}}$ and magnitude m_{CModel} . At a fixed magnitude, the distribution is modeled as a mixture of unresolved and resolved components which, with appropriately chosen priors for the star-to-galaxy number ratio, allows us to compute p_S , the posterior probability that a source is unresolved. We apply this method to the Rubin Data Preview 2 object catalog and validate the resulting classifications using external HST labels in the COSMOS field. Using these labels as the ground truth, we quantitatively compare the performance of the p_S -based classifier with that of the Rubin **extendedness** parameter. We find that the p_S -based classifier consistently outperforms the Rubin **extendedness** baseline, with the improvement increasing as additional bands are incorporated. Approximately, the p_S classifier achieves the same classification performance at a magnitude limit about half a magnitude fainter than the **extendedness** classifier. Depending on the scientific application, this morphological classification can be further improved by incorporating complementary color-based classifiers.

Keywords: Galaxies (573) — Cosmology (343) — Stellar astronomy (1583) — Solar physics (1476)

1. INTRODUCTION

The morphological classification of resolved and unresolved sources⁴, commonly referred to as “star-galaxy separation”, is a fundamental step in the analysis of wide-field imaging surveys such as the Rubin’s Legacy Survey of Space and Time (LSST). At bright magnitudes this problem is relatively simple, since stars outnumber galaxies and galaxies that exist are obviously extended on the sky. Star-galaxy separation becomes

Corresponding author: Željko Ivezić

Email: ivezic@uw.edu

⁴ Note that quasars and galaxies can be unresolved and even some stars can be resolved, such as Asymptotic Giant Branch stars.

increasingly harder at faint magnitudes, such as those probed by Rubin’s Legacy Survey of Space and Time (LSST; for details see [Ž. Ivezić et al. 2019a](#) and references therein) because galaxy counts rapidly increase with magnitude, while their angular sizes become smaller. The selection of clean faint stellar samples is especially hard.

A variety of methods have been proposed in the literature for morphological classification of sources in images. [C. T. Slater et al. \(2020\)](#) (hereafter SIL2020) discussed the statistical foundations of morphological star–galaxy separation and showed that many of the star–galaxy separation metrics in common use today, e.g., by Sloan Digital Sky Survey (SDSS) *photo* pipeline ([R. H. Lupton et al. 2002](#)), or SExtractor pipeline ([E. Bertin & S. Arnouts 1996](#)), are closely related both to each other, and to the Bayesian model odds ratio, first derived by [W. L. Seaborg \(1979\)](#). SIL2020 quantified the scaling of various algorithms with the source noise properties and showed that the Bayesian model odds ratio achieves the best performance. They also showed how to probabilistically combine multiple measurements, either of the same type (e.g., subsequent exposures), or differing types (e.g., multiple bandpasses), or differing methodologies (e.g., morphological and color-based classification).

Following SDSS, LSST photometric catalogs use a binary classifier `extendedness` in each band, based on the threshold difference between PSF, m_{PSF} , and CModel, m_{CModel} , magnitudes, $\Delta m = m_{\text{PSF}} - m_{\text{CModel}}$. For unresolved sources, this Δm is vanishing within noise, with `extendedness`=0, while for resolved objects, selected by $\Delta m > 0.016$ mag, `extendedness`=1. Resolved objects have positive Δm because their PSF flux is smaller than CModel flux due to a narrow weighting PSF profile. For an illustration of Δm distribution in Rubin’s Data Preview 1 catalogs, see Figure 28 in [Vera C. Rubin Observatory Team et al. \(2026\)](#) or Figure 3 further below).

Although `extendedness` was successfully used in thousands of papers based on SDSS catalogs, this approach is not optimal for LSST. As discussed in SIL2020, when the star–galaxy separation threshold is optimized at the faint, low S/N, end of the data, the ability to recognize barely resolved objects at the bright end is not fully exploited. Furthermore, the binary nature of `extendedness` parameter prevents optimal combination of information from different bands, as well as from other classification methods (e.g., based on colors, or other datasets). SIL2020 showed how to turn Δm into a probability distribution, which then allows proper probabilistic combination of multiple bands. With appropriately chosen priors for star-to-galaxy counts ratio, the posterior probability that a source is unresolved (or resolved), can be computed in Bayesian framework.

The main goal of this paper is to implement the steps towards Bayesian star–galaxy separation outlined in SIL2020 using Rubin’s Data Preview 2 data. We describe our dataset in §2, and in §3 discuss star–galaxy separation methodology, including quantification of its performance using HST-based labels. We discuss our results and how to retrieve our catalog with improved star–galaxy separation parameters in §4.

2. DATASETS

Here we describe two principal datasets used in our analysis: Rubin Data Preview 2 catalogs and HST data with star–galaxy labels used for testing methods discussed in §3.

2.1. Rubin Data Preview 2

All data collected with the LSST Camera [Vera C. Rubin Observatory Team \(2026\)](#) during the commissioning of the Rubin Observatory will be released as Data Preview 2⁵ (hereafter DP2). Coadded deep images and resulting catalogs have been released in July 2026 as Early Data Preview 2. These coadded images are based on about 24,000 single-visit images and cover about 3,000 sq. deg. of sky.

In our analysis we focus on the COSMOS field because it is both already well observed by Rubin (to only about 1 magnitude shallower limit than anticipated for 10-year LSST), and space-based HST COSMOS2020/FARMER star and galaxy labels are available ([J. R. Weaver et al. 2022](#)). As a control sample for testing code robustness, we also used Rubin data for the Extended Chandra Deep Field South (ECDFS). The Rubin $5 - \sigma$ depth for point sources in the COSMOS field, estimated as the median magnitude of point sources with photometric errors of ~ 0.2 mag, is (25.66, 26.66, 26.36, 25.91, 24.94, 23.59) in the *ugrizy* bands, respectively.

⁵ See <https://dp2.lsst.io/>

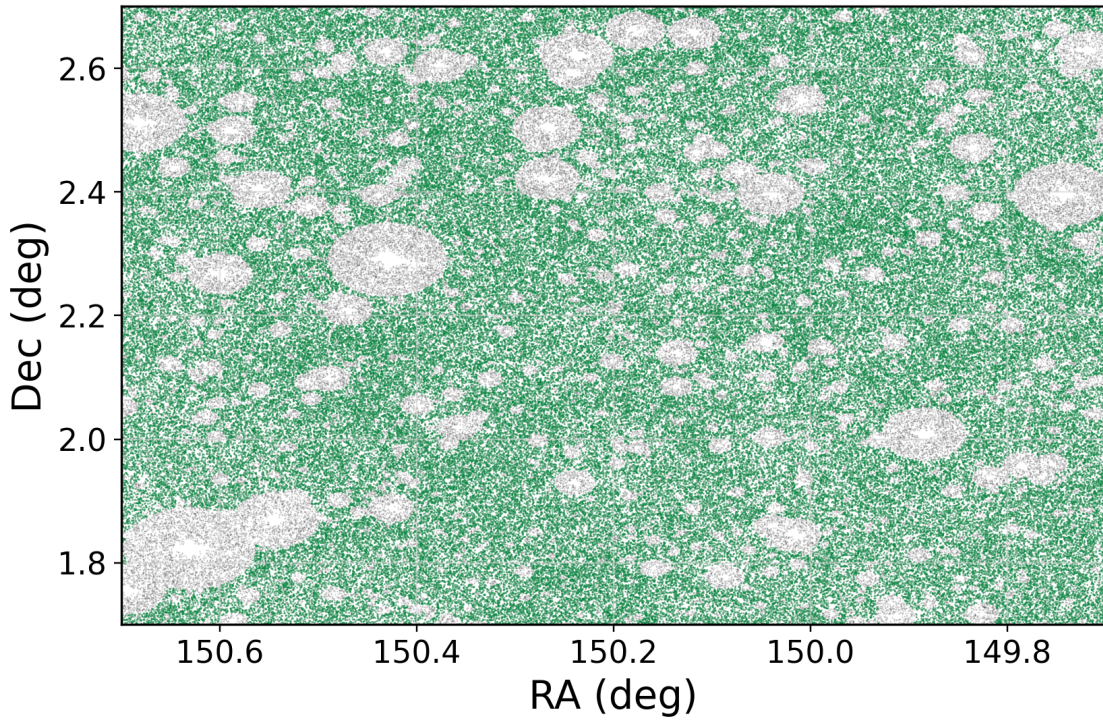


Figure 1. The footprint for the DP2-COSMOS sample analyzed here. Gray points show Rubin DP2 objects and green points show the subset matched to the HST COSMOS2020/FARMER sample a 0.5 arcsec matching radius. The white regions in the centers of gray areas correspond to areas where no objects are listed in either catalog.

2.2. HST Star-galaxy Classification Labels in the COSMOS Field

For the COSMOS validation sample, we select Rubin DP2 sources in the rectangular region $149.7 < \text{RA} < 150.7$ and $1.7 < \text{Dec} < 2.7$, with $16 \leq r_{\text{CModel}} < 26$. Objects listed in the COSMOS2020/FARMER reference catalog are matched to DP2 sources using a 0.5 arcsec matching radius. Within the selected region and magnitude ranges, the DP2 sample contains 269,606 objects, of which 193,586 are matched to the COSMOS2020/FARMER objects and 76,020 are unmatched (a matching fraction of 72%). This somewhat low matching fraction is due to large areas around bright stars that are masked in the COSMOS2020/FARMER catalog (see Figure 1).

The matched validation sample contains 8,788 stars and 184,798 galaxies, classified using HST’s `truth_binary = 1` for stars and `truth_binary = 0` for galaxies. Their number counts as function of magnitude are shown in Figure 2. The low fraction stars is due to very faint sample limit $r \sim 26$) and high Galactic latitude ($b \sim 50^\circ$).

2.3. The color-magnitude distributions of sources classified using HST labels

The external COSMOS2020/FARMER labels provide the reference classification, considered here as ground truth, for examining the distribution of stars and galaxies in various diagrams and for measuring the performance of star-galaxy separation methods. Nevertheless, it is likely that some very faint sources classified by HST as stars might be unresolved galaxies even at HST’s resolution. Therefore, the HST labels will help us assess whether a source should be resolved or unresolved in Rubin images, but do not necessarily provide definitive information about astrophysical nature of a source (e.g., to separate unresolved stars and galaxies, one would need spectra or broad-band photometry over a much wider wavelength range, say, in near-infrared K band). Furthermore, no classification is perfect and thus we first examine the distribution of HST-classified “stars” and “galaxies” in various informative color-color diagrams constructed with Rubin photometry.

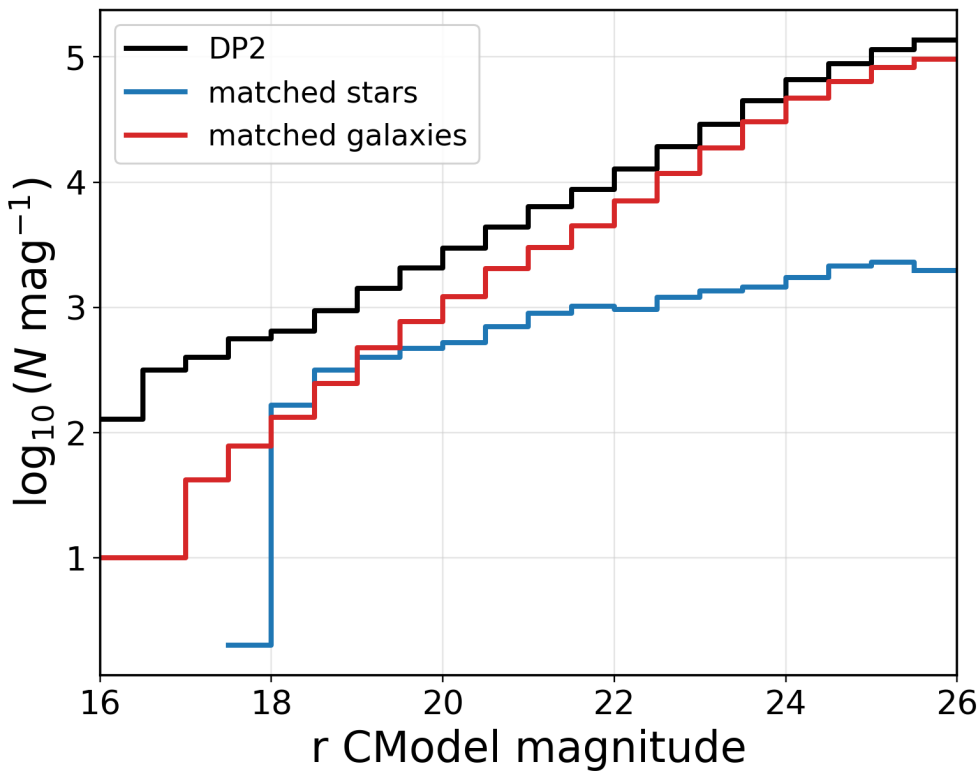


Figure 2. Magnitude distribution of the DP2-COSMOS sample. The black curve shows all selected DP2 sources, while the blue and red curves show matched COSMOS2020/FARMER sources classified as stars and galaxies, respectively. Counts are shown per magnitude, using 0.5 mag wide bins. Note that galaxies outnumber stars at the faint end by more than an order of magnitude.

The top left panel in Figure 3 shows the distribution of stars and galaxies classified using HST labels in the $\Delta m = m_{\text{PSF}} - m_{\text{CModel}}$ vs. m_{CModel} diagram. As we describe in detail in the next Section, modeling this distribution is a key ingredient of the Bayesian star-galaxy separation method presented here.

The top right panel in Figure 3 shows that faint galaxies ($r > 24$), in addition to outnumbering stars at faint magnitudes, have on average bluer $g - i$ colors than stars. The clearly different distributions of stars and galaxies in color-color diagrams shown in the bottom two panels in Figure 3 demonstrate that colors can be used to enhance morphological star-galaxy separation. Nevertheless, Figure 4 and the top left panel in Figure 3 clearly show that increased photometric noise close to the faint magnitude limit will cause deteriorated performance for both morphological and color-based star-galaxy separation methods. Extending the start of this deterioration to a fainter magnitude limit is one of the primary goals of this work. It is notable that the top right panel in Figure 4 implies that most unresolved faint blue sources have $u - g$ color more typical of low-redshift quasars than of main sequence stars (for a detailed discussion of the blue star counts see L. Palaversa et al. 2026).

2.4. The color-magnitude distributions of sources classified using *extendedness* parameter

Difficulties with star-galaxy separation encountered at the faint end are illustrated in Figure 5, which compares subsamples classified using HST ground-truth labels and the default Rubin *extendedness* binary class in the r band. As evident, a significant fraction of faint sources within about two magnitudes from the faint limit are misclassified when using default *extendedness* parameter. Misclassifications have a more severe impact on stellar sample: only 26% of “stars” selected by *extendedness*=1 condition are unresolved in HST data. This high level of stellar sample contamination by faint galaxies increases towards the faint end (see the bottom left panel in Figure 5). As the same time, the selection completeness for “stars” (objects unresolved by HST) is only 67%. Due to much higher counts at faint magnitudes and high Galactic latitudes,

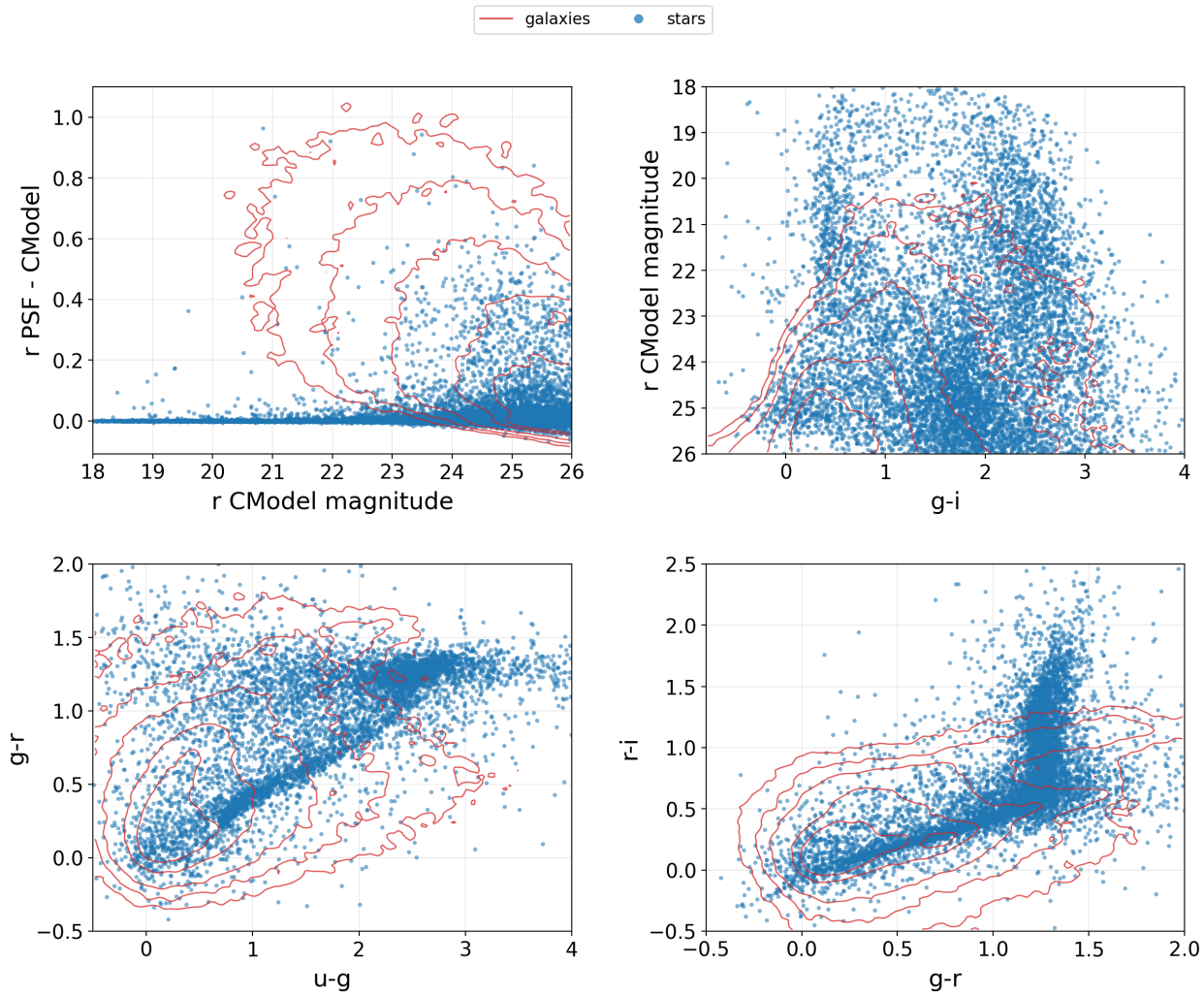


Figure 3. The distribution of stars (blue dots) and galaxies (red contours), classified using HST labels, in various informative diagrams. The CModel-magnitude based colors are corrected for the ISM dust extinction, while the r -band CModel magnitude is not. The top left panel shows the $\Delta m = m_{\text{PSF}} - m_{\text{CModel}}$ vs. m_{CModel} diagram, where stars form a narrow sequence centered on $\Delta m \approx 0$ and for galaxies $\Delta m > 0$. The top right panel shows the r vs. $g - i$ color magnitude diagram and two bottom panels show most informative color-color diagrams. The main stellar locus, dominated by main sequence stars and red giants, is clearly visible with its blue end redwards from $u - g \sim 0.6$; unresolved sources with $u - g < 0.6$ are expected to be a mixture of low-redshift quasars, white dwarfs and unresolved binaries.

the purity and completeness for galaxies are much higher: 98% and 89%, respectively. In other words, a simple classifier that declares every object to be a galaxy would formally perform quite well. Therefore, we will focus our analysis in the next Section on the classifier’s performance when selecting unresolved objects.

2.5. Suspect HST Classification Labels

There is a subset of objects classified as “stars” (unresolved) by HST which appear clearly resolved when using Rubin’s default `extendedness` parameter. Figure 6 shows their distribution in color-color diagrams, with symbols color-coded by Δm in the r band. As evident, their color-color distribution is inconsistent with main stellar locus and similar to that expected for barely resolved or unresolved binary stars (see Figures 2 and 4 in V. Smolčić et al. 2004.) To test this hypothesis, we visually inspected several hundred images (443 objects with HST label “star”, $r < 23$ and $\Delta m > 0.05$ mag; see Figure 7 for an example). Only 1.4% of selected objects are detected as “isolated” (computed using processing flag “detect_isolated”) and about

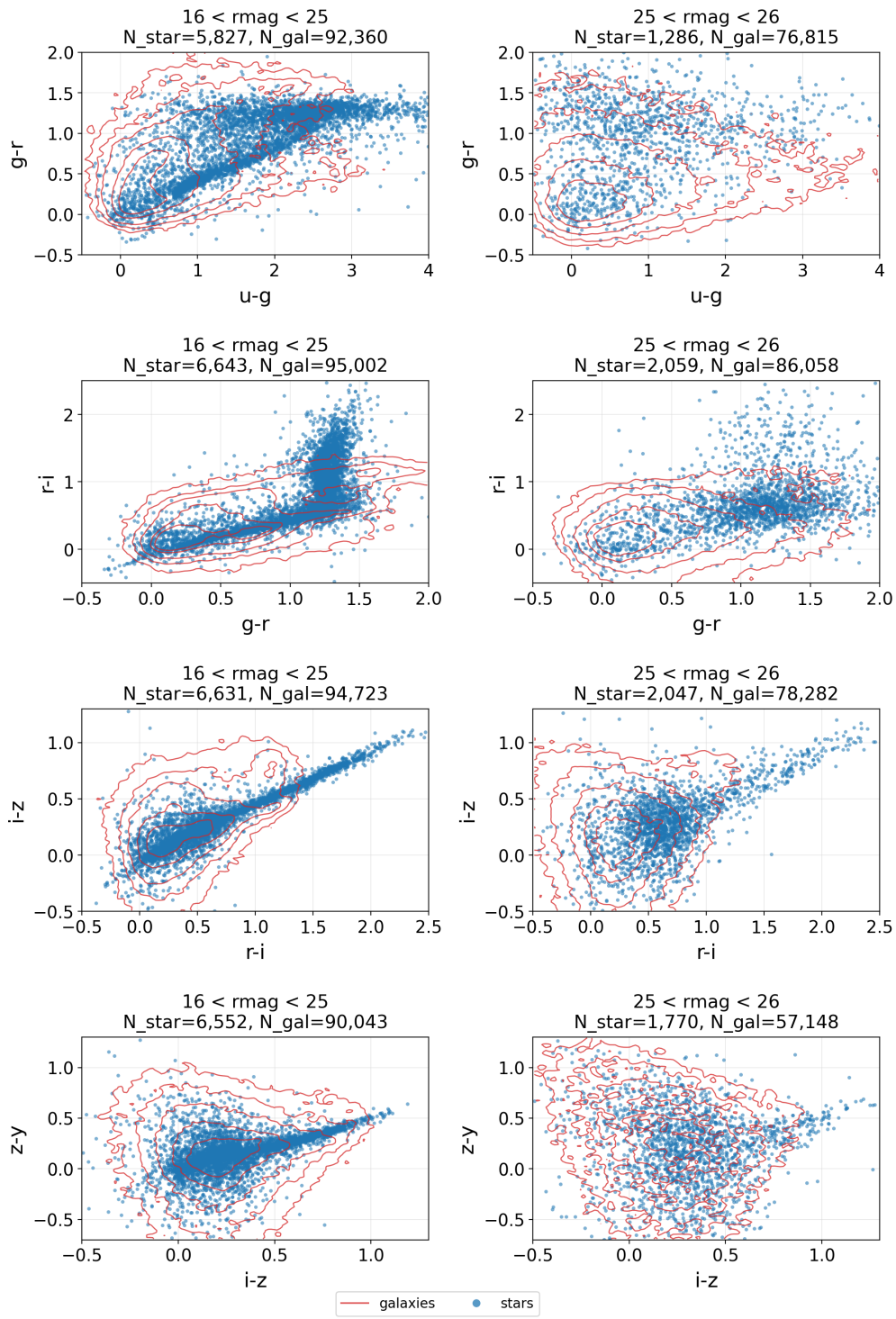


Figure 4. Similar to the bottom two panels in Figure 3, except that here all principal color-color diagrams are shown and the data are split by the r -band CModel magnitude to show deterioration of photometry close to the faint magnitude limit (left: $16 < r_{\text{mag}} < 25$; right: $25 < r_{\text{mag}} < 26$). The $5 - \sigma$ depth for point sources in the r band for this dataset is 26.36.

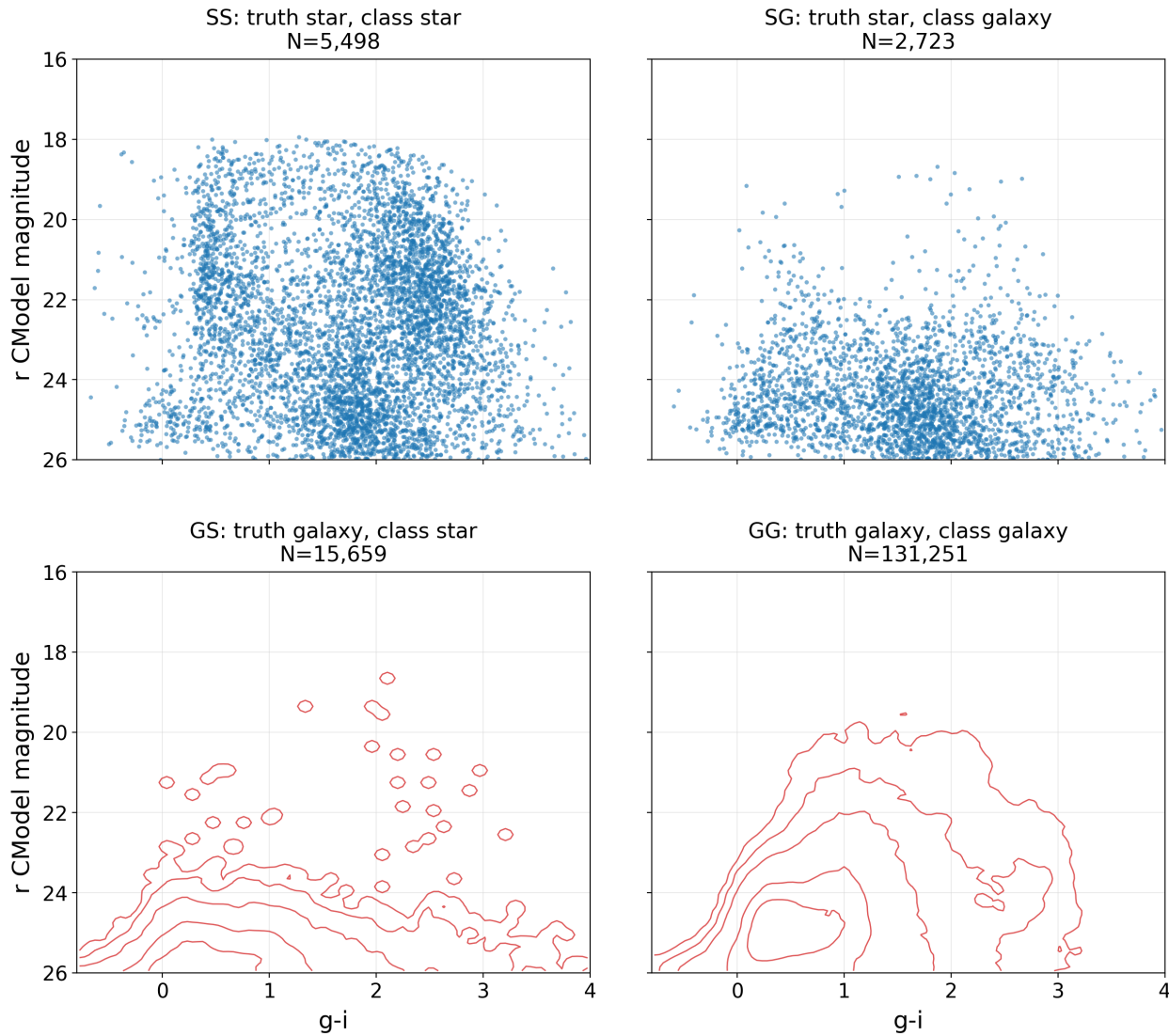


Figure 5. Performance of Rubin’s default **extendedness** parameter in the r band relative to HST ground-truth labels. The top left and bottom right panels show correctly classified unresolved and resolved sources in the r vs. $g - i$ color-magnitude diagram, with counts shown in the title above each panel. The bottom left and the top right panels show misclassified sources. Note that the stellar sample selected by **extendedness**=0 is severely contaminated by galaxies for $r_{\text{CModel}} > 24$ (bottom left panel).

half seem to have a nearby stellar-like companion. In some cases a bright source or an extended galaxy is nearby, and about one tenth or fewer appear bluish and consistent with quasars. Detailed examination of such cases is beyond the scope of this work.

Because we aim to test statistical performance of our star-galaxy separation method on unambiguous objects, we exclude objects with HST “star” label that both appear resolved in Rubin data and also have Rubin colors inconsistent with the main stellar locus.

DANIEL: can you please add a paragraph here about how you rejected some HST stars by colors and what were the numbers (fractions) of rejected objects (and how you handled variation with magnitude)?

3. STAR-GALAXY SEPARATION METHODOLOGY

We first briefly review the Bayesian star-galaxy separation method proposed in SIL2020 and then apply it to single-band Rubin data, followed by its extension to multi-band case. Our principal measured object property is the difference between PSF, m_{PSF} , and CModel, m_{CModel} , magnitudes, $\Delta m = m_{\text{PSF}} - m_{\text{CModel}}$.

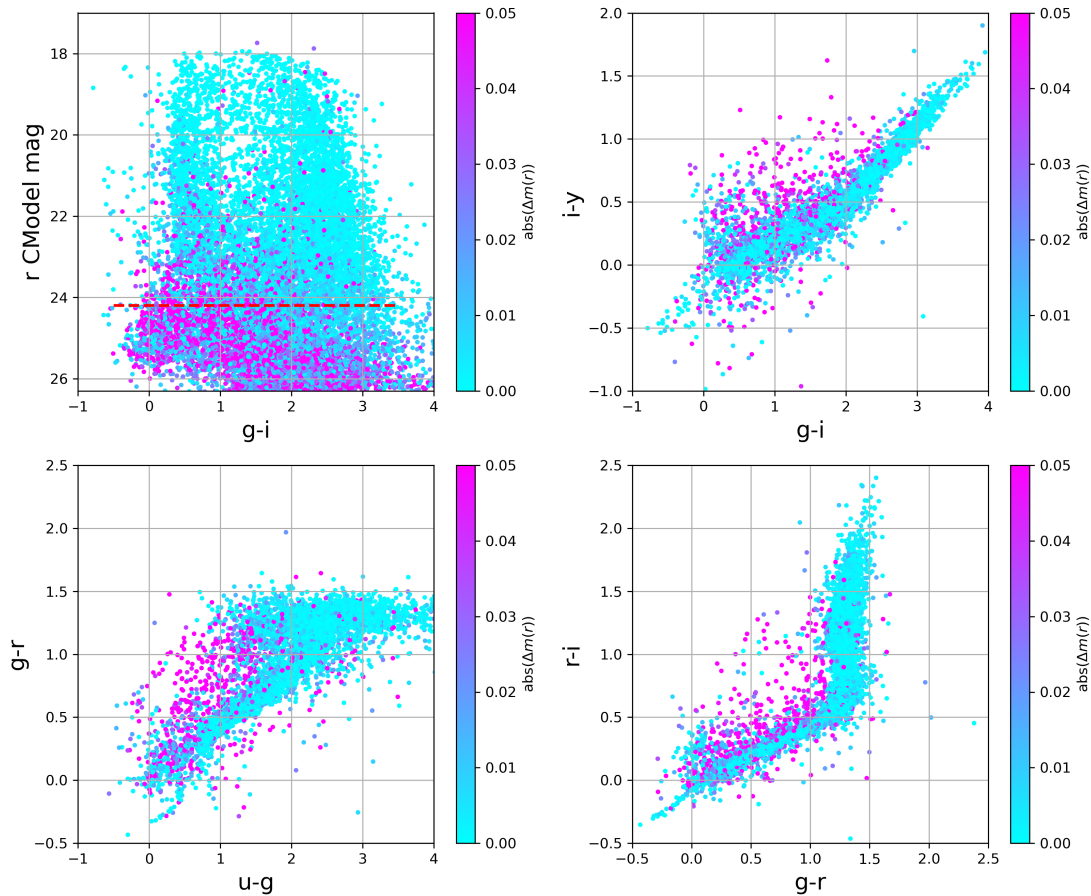


Figure 6. The distribution of objects classified as “stars” by HST, but which appear clearly resolved when using Rubin’s default `extendedness` parameter, in color-magnitude (top left) and informative color-color diagrams. Only objects with $r < 24.2$ are shown in the three color-color diagrams. Symbols are color-coded by the absolute value of Rubin’s Δm in the r band. The color-color distribution of objects with $|\Delta m| > 0.05$ mag (about 3% of all objects classified by HST as “stars” and brighter than $r = 23$) is inconsistent with main stellar locus and similar to that expected for barely resolved or unresolved binary stars.

For unresolved sources, this Δm is vanishing within noise, while for resolved objects $\Delta m > 0$ (see Figure 3). A comparison and statistical discussion of this quantity and several other quantities proposed for star-galaxy separation in the literature is available in SIL2020.

3.1. Bayesian Star-Galaxy Separation Method from SIL2020

We first focus on the single-band case and then will extend our discussion to then multi-band case. SIL2020 showed using realistic simulations that adopting a fixed pre-defined threshold for Δm in order to select resolved objects is not optimal. Not only that this threshold should depend on magnitude due to variation of object populations, signal-to-noise ratio due to statistical noise, and sky location due to varying count ratio of galaxies and stars, but for optimal combination of information from different bands, or from other classification methods (e.g., based on colors), observed Δm distribution needs to be transformed into a proper probability density function.

The subsample of objects with HST labels can be used to quantify Δm distribution separately for unresolved and resolved objects. However, such a “training” sample is currently available only for a very small fraction of LSST footprint (less than 1% of the area). Instead, we use the fact that very large numbers of objects that remain even after binning in both sky location and magnitude can robustly constrain two

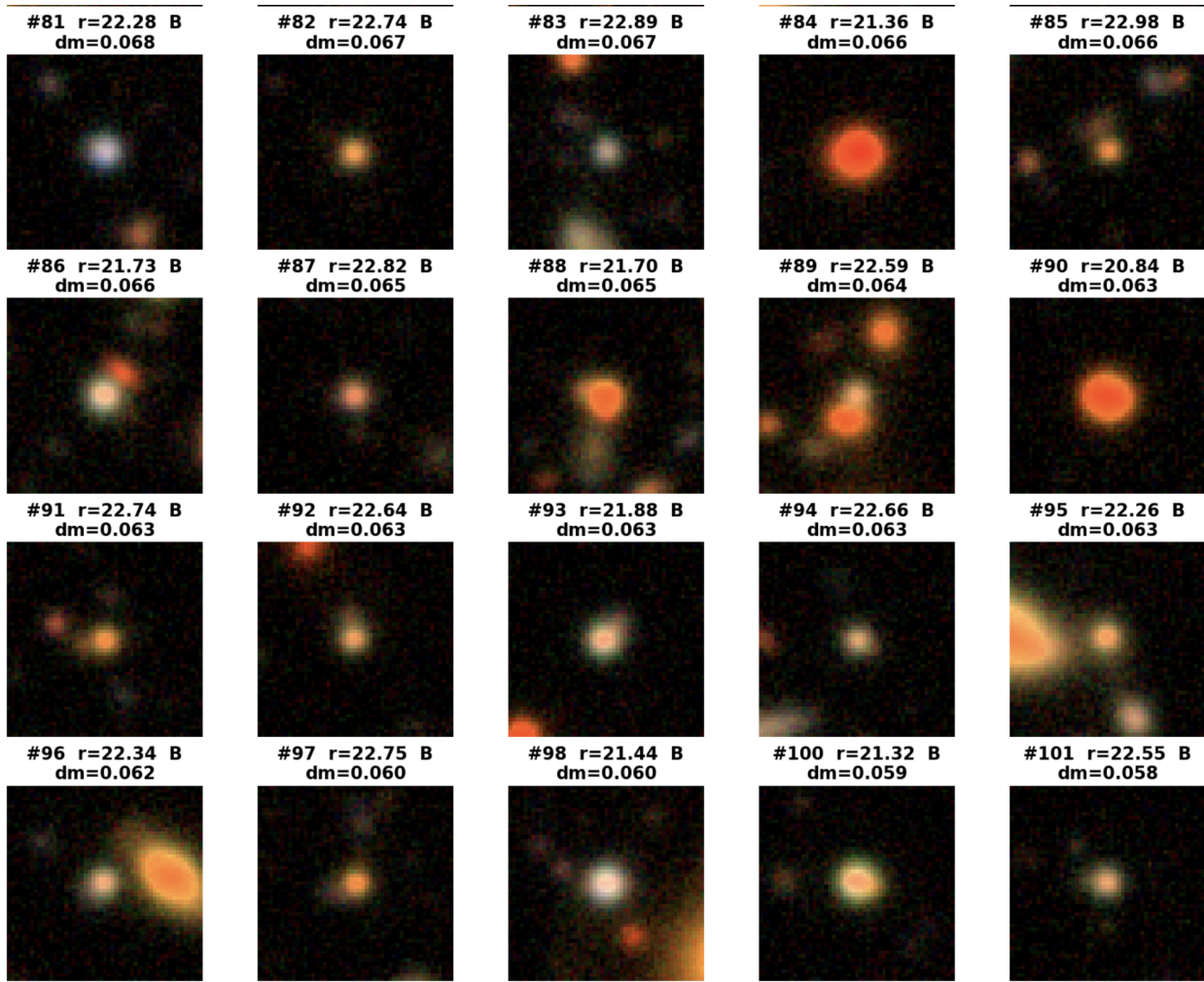


Figure 7. The Rubin *gri* color composite images extracted from “deep_coadd”, 10 arcsec by 10 arcsec (50x50 pixels), for a random subsample of 443 objects selected as “stars” in HST and resolved in Rubin, and brighter than $r = 23$. About 98.6% are detected as blended objects.

components of Δm distribution even when such labels are not available a priori. We only use the subsample of objects with HST labels to independently assess the performance of our method.

3.2. Mixture Model for Δm vs. m_{CModel} diagram

We assume that a sample is selected from a judiciously chosen sky region over which the variation of stellar and galaxy population can be assumed statistically uniform. Over angular scales of several degrees, stellar counts vary much more rapidly than galaxy counts (large galaxy clusters are an exception but not important for subsequent discussion). Based on the analysis of state-of-the-art Milky Way stellar population models by L. Palaversa et al. (2025), a sky region of several square degrees, corresponding to Rubin tracts⁶ is sufficiently small for this purpose (and consistent with our choice of the COSMOS field for quantitative analysis).

Given a “tract” catalog, and at a fixed CModel magnitude (see below for binning details), the distribution of Δm in a given band (not explicitly marked for notational simplicity) is modeled as a mixture of an

⁶ In the Rubin Science Pipelines, tracts and patches are the two levels of spatial subdivision used for coadded images and object catalogs. They are designed to make processing and storage scalable while preserving continuity across the sky. A tract is a relatively large region of the sky, approximately $1.75^\circ \times 1.75^\circ$ (the exact size depends on the sky map projection). that is processed as a unit. Tracts overlap slightly with neighboring tracts to avoid edge effects. Each tract is subdivided into a regular grid of patches, with each tract in DP2 containing 81 patches. The patch size is about 12 arcmin by 12 arcmin (approximately, one CCD).

unresolved (star) component and a resolved (galaxy) component,

$$p(\Delta m | m) = w_U f_{\text{unr}}(\Delta m) + w_R f_{\text{res}}(\Delta m), \quad (1)$$

where $w_U + w_R = 1$, and f_{unr} and f_{res} are properly normalized probability density functions. With this decomposition, the probability that an object is unresolved is given by Bayes' theorem as

$$p_S(\Delta m) = p(\text{unr} | \Delta m) = \frac{w_U f_{\text{unr}}(\Delta m)}{w_U f_{\text{unr}}(\Delta m) + w_R f_{\text{res}}(\Delta m)}. \quad (2)$$

and the probability that it is resolved is $p_G = 1 - p_S$ (note that we use indices S and unr , or G and res interchangeably).

The same model predicts the performance, the completeness and contamination for each class, as function of a hard Δm threshold. If $\Delta m < \Delta m_{\text{cut}}$ is classified as unresolved,

$$\text{Completeness}_S = \int_{-\infty}^{\Delta m_{\text{cut}}} f_{\text{unr}}(\Delta m) d\Delta m, \quad (3)$$

and

$$\text{Contamination}_S = \frac{w_U \int_{-\infty}^{\Delta m_{\text{cut}}} f_{\text{unr}}(\Delta m) d\Delta m}{w_U \int_{-\infty}^{\Delta m_{\text{cut}}} f_{\text{unr}}(\Delta m) d\Delta m + w_R \int_{-\infty}^{\Delta m_{\text{cut}}} f_{\text{res}}(\Delta m) d\Delta m}. \quad (4)$$

We test these predictions below using HST-based labels.

Therefore, a good model for $p(\Delta m | m)$ (eq. 1) is crucial. Informed by observed Δm distributions, we model the unresolved component, f_{unr} as a single Gaussian with mean μ_U , width σ_U (and weight w_U). The resolved component is modeled as a sum of K log-normal distributions with weight $w_{R,k}$, location $\mu_{R,k}$, scale $\sigma_{R,k}$, and shape parameter $\alpha_{R,k}$. The full mixture probability density function is

$$p(\Delta m | m) = w_U \mathcal{N}(\Delta m | \mu_U, \sigma_U) + \sum_{k=1}^K w_{R,k} \mathcal{LN}(\Delta m | \mu_{R,k}, \sigma_{R,k}, \alpha_{R,k}), \quad (5)$$

where $\mathcal{N}$ denotes the Gaussian and $\mathcal{LN}$ the log-normal distribution, and the weights are constrained to sum to unity: $w_U = 1 - \sum_k w_{R,k}$ (functions $\mathcal{N}$ and $\mathcal{LN}$ are properly normalized probability density functions). The galaxy-to-star count ratio, which serves as Bayesian prior, is equal to

$$N_{GS} \equiv \frac{N_G}{N_S} = \frac{\sum_{k=1}^K w_{R,k}}{w_U}. \quad (6)$$

The expression given by eq. 2 can be recast as

$$p_S(\Delta m) = \frac{1}{1 + N_{GS} BF}, \quad (7)$$

where Bayes factor is equal to

$$BF = \frac{f_{\text{res}}}{f_{\text{unr}}}. \quad (8)$$

With $N_{GS} \sim 30$ (see Figure 8), it is sufficient to have $BF \sim 0.033$ for p_S to drop below 0.5 (because the sample is highly imbalanced). This decomposition in terms of prior and Bayes factor will be important later when combining multiple sources of classification information.

3.2.1. Fitting Details

The fitting is performed independently in bins of CModel magnitude. At bright magnitudes ($m_{\text{CModel}} < 22$), where the resolved population is relatively simple, we use $K = 2$ resolved components in bins of width 1–3 mag spanning $15 < m_{\text{CModel}} < 22$. At faint magnitudes ($m_{\text{CModel}} \geq 22$), where the galaxy morphology distribution is more complex, we use $K = 3$ resolved components in finer bins of width 0.5–1 mag spanning

22 $\leq m_{\text{CModel}} < 27$. Within each magnitude bin, Δm values in the range $(-0.5, 1.5)$ are binned into 300
 23 bins and normalized to unity. The mixture model is fit to these histograms using bounded least-squares
 24 optimization with Poisson-like uncertainties $\sigma_i = \sqrt{N_i + 1} / (N \delta)$, where N_i is the count in bin i , N is the
 25 total number of sources in the magnitude slice, and δ is the histogram bin width.

26 This fitting procedure is carried out in three steps. In the first step, all mixture parameters are fit
 27 independently in each magnitude bin. This free fit works well at bright magnitudes, where unresolved and
 28 resolved sources are clearly separated in Δm . However, at faint magnitudes ($m_{\text{CModel}} \gtrsim 24$), photometric
 29 noise broadens the unresolved Gaussian until it becomes degenerate with the resolved components. The
 30 fitter can no longer reliably distinguish the two populations, causing the unresolved Gaussian to absorb flux
 31 from the galaxy distribution. This inflates the inferred star fraction at faint magnitudes.

32 To prevent this degeneracy, we impose smooth constraints on the unresolved component. In the second
 33 step, the per-bin star parameters from step 1 are replaced with smooth functions of magnitude: μ_U is fixed
 34 to a single value (the median across all bins, typically ≈ 0 in each band); $\sigma_U(m)$ is fit with the photometric
 35 error model from B. Sesar et al. (2007) (their eq. 5),

$$\sigma_U(m) = \sqrt{\sigma_0^2 + a^2 \cdot 10^{b(m-25)}}, \quad (9)$$

36 where σ_0 , a , and b are free parameters; and $w_U(m)$ is modeled as a linear fit to $\log_{10}(w_U)$ versus magnitude,
 37 using only bins brighter than $m_{\text{CModel}} = 24$ where the free fit is reliable. This smooth model is then
 38 extrapolated to fainter magnitude bins. In the third step, with the unresolved component (μ_U , σ_U , w_U) fixed
 39 to the smooth curves from step 2, only the resolved log-normal parameters are re-fit in each magnitude bin.
 40 The resolved weights are rescaled to sum to $(1 - w_U)$ so that the total mixture remains a proper probability
 41 density. The effectiveness of this constrained fitting procedure is demonstrated in Figure 8, which compares
 42 the galaxy-to-star count ratio $\log_{10}(N_{\text{gal}}/N_{\text{star}})$ as a function of magnitude for models with and without the
 43 smooth constraints. The unconstrained model exhibits a spurious turnover at faint magnitudes, while the
 44 constrained model recovers the expected monotonically increasing ratio in agreement with the COSMOS2020
 45 external truth labels (note that in this sample at the faint end galaxies outnumber stars by more than a
 46 factor of 30).

47 The model parameters are looked up by the CModel magnitude bin in which each source falls. The quality
 48 of the fits is illustrated in Figure 9, which shows the mixture model overlaid on the observed Δm histograms
 49 in each magnitude bin, and in Figure 10, which shows the 2D data, model, and residual maps in the Δm vs.
 50 m_{CModel} plane.

3.3. The Performance of the Δm Mixture Model

51 Here we use HST labels to evaluate completeness and contamination for the stellar sample, and to compare
 52 them to mixture model predictions. We focus on the stellar subsample and the “transition” magnitude range:
 53 $24 < r < 24.5$: brighter than this limit the population mixing is much smaller, while fainter than this limit
 54 the performance deteriorates rapidly (especially for the stellar subsample).

3.4. The Multi-band Case

55 We expect classification to improve with more data but combining the information from different bands is
 56 statistically complex problem...

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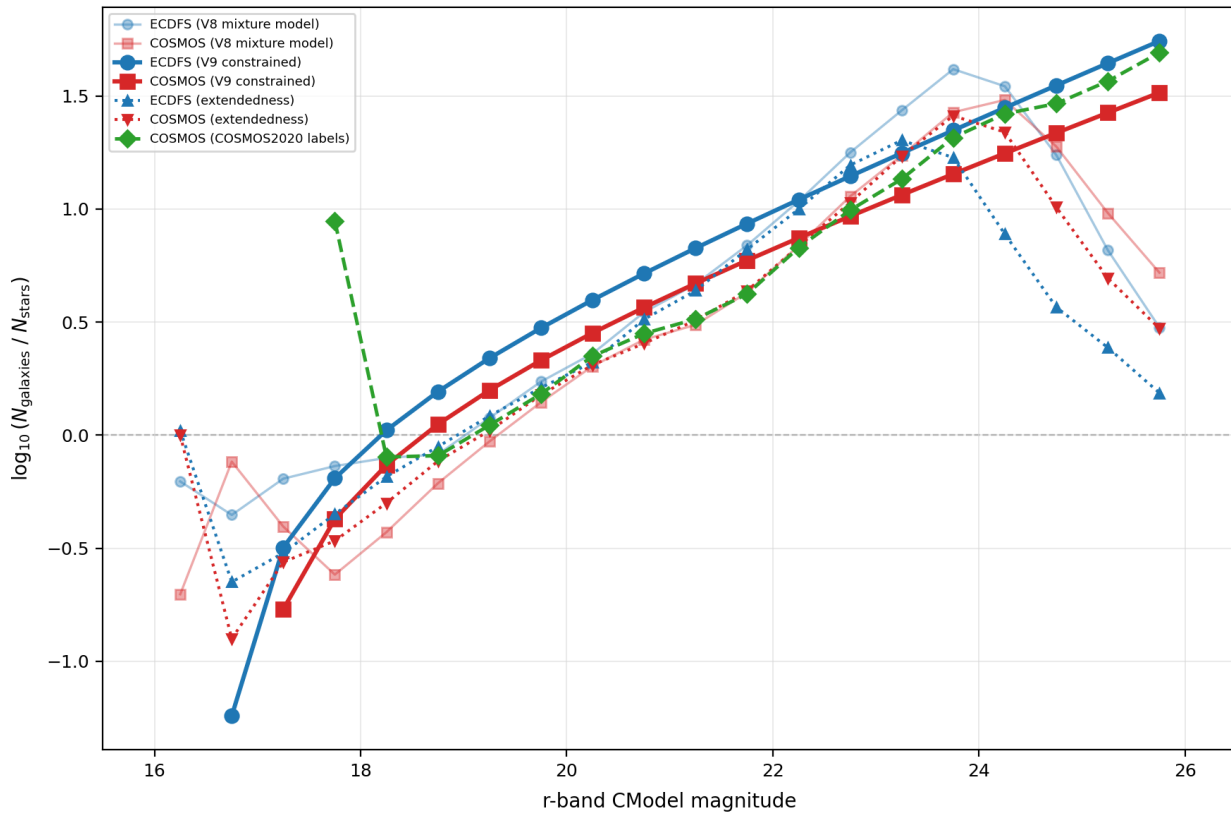


Figure 8. Galaxy-to-star count ratio $\log_{10}(N_{\text{gal}}/N_{\text{star}})$ vs. r-band CModel magnitude. The unconstrained mixture model (marked V8, faded curves) exhibits a spurious turnover at $m \gtrsim 24$ due to the unresolved Gaussian absorbing galaxy flux. The constrained model (marked V9, bold curves) recovers the expected monotonically increasing ratio, in good agreement with the COSMOS2020 truth labels (green dashed). Dotted lines show the Rubin **extendedness**-based classification for comparison (which also shows spurious turnover at the faint end).

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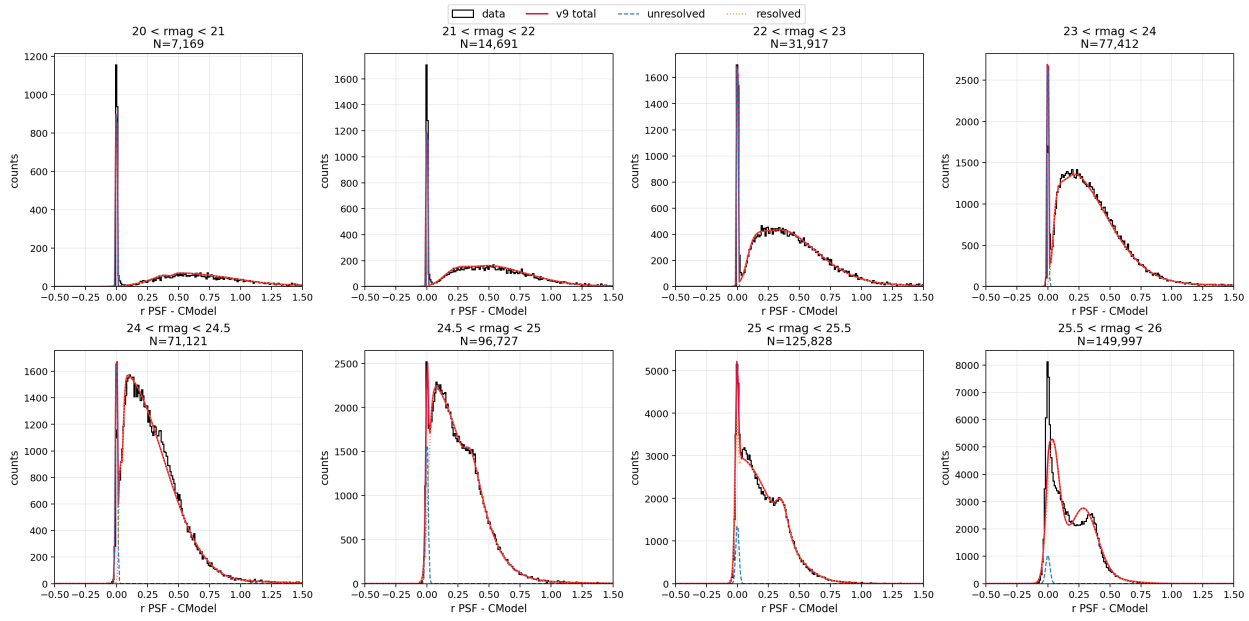


Figure 9. Observed Δm histograms (black) and fitted mixture models (red) for the r-band in each CModel magnitude bin. The unresolved Gaussian component is shown as a blue dashed line and the resolved skew-normal components as orange dotted lines.

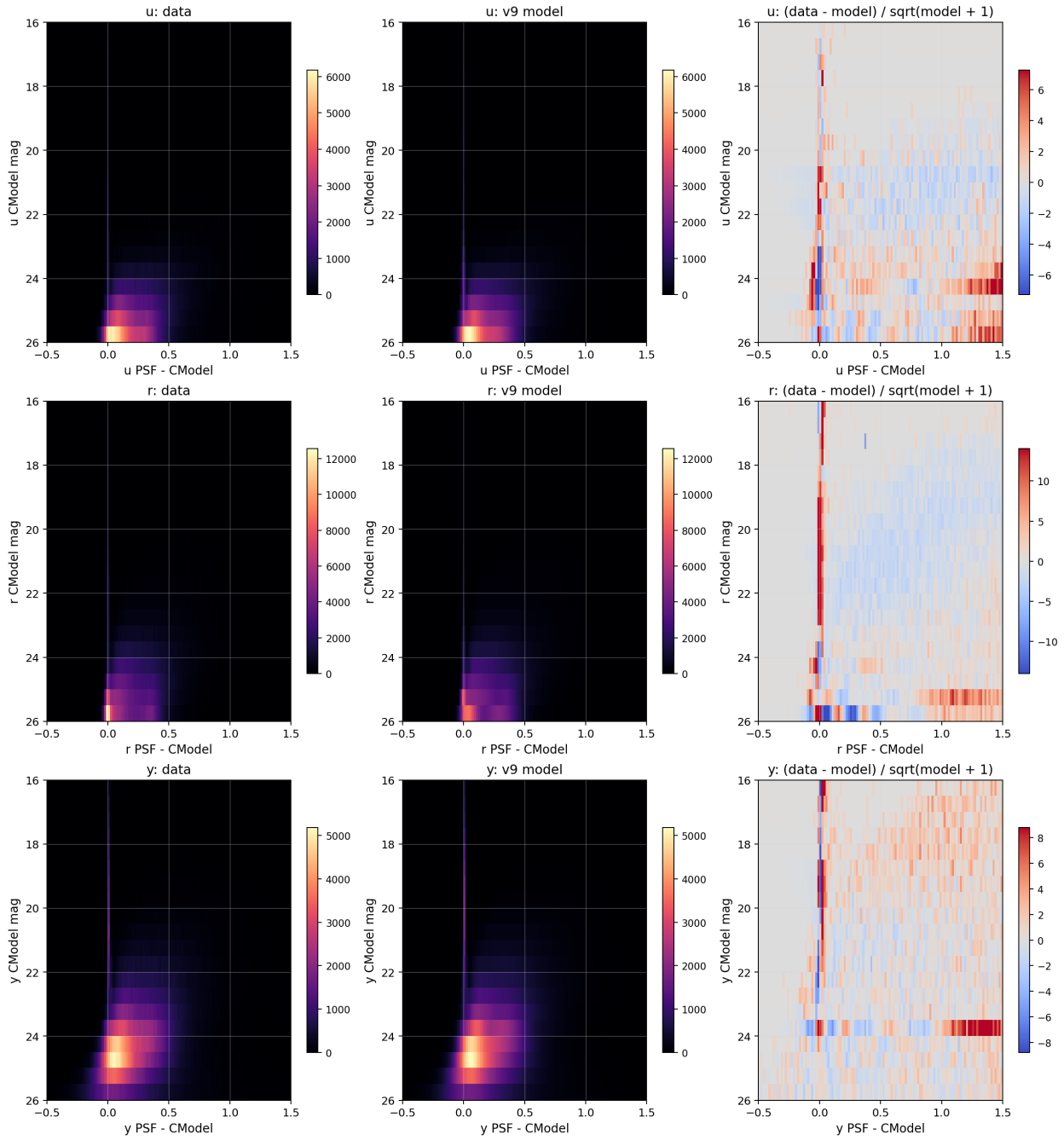


Figure 10. Two-dimensional comparison of the observed (left), modeled (center), and residual (right) source density in the Δm vs. m_{CModel} plane. The small residuals demonstrate that the constrained mixture model provides an adequate description of the data across the full magnitude range.